

Lab 3 - Routing Protocol

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SECTION: 01

Introduction

You are given a Packet Tracer file, which requires some work IP addressing and routing protocol configuration. You must follow all the steps carefully and answer the given questions.

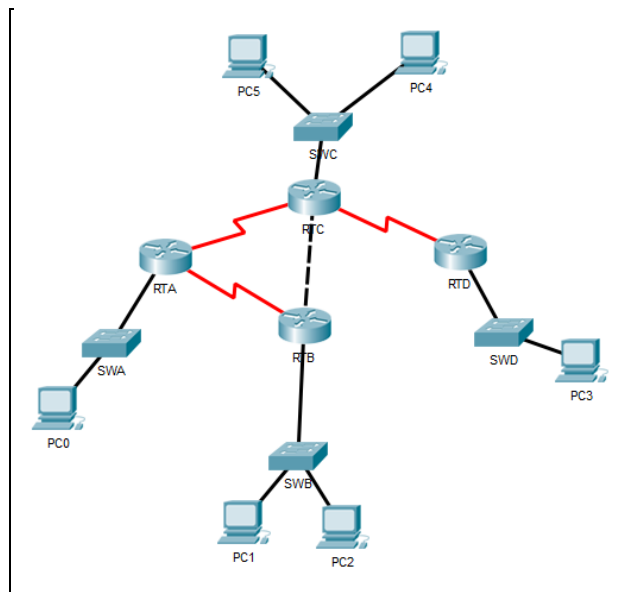


Figure 1

Task 1: IP addressing

Step 1: Fill in Table 1 below with the correct information. Note: The information may be found under the **Config** tab of each router (refer to Figure2).

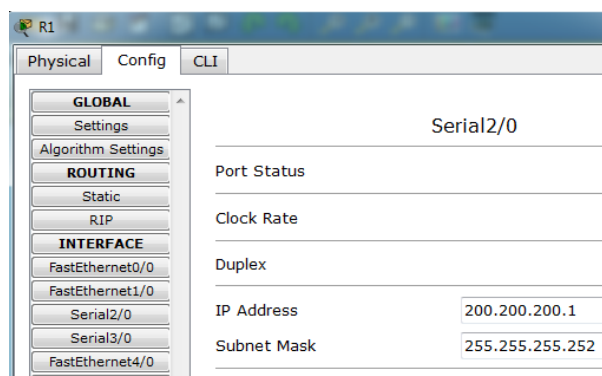


Figure 2

Table 1

#	Device Name	Interface	IP Address	Subnet Mask
1	RTA	Se2/0	172.16.230.5	255.255.255.252
2		Se3/0	172.16.230.1	255.255.255.252
3		Fa0/0	172.16.224.255	255.255.254.0
4	RTB	Se2/0	172.16.230.2	255.255.255.252
5		Fa0/0	172.16.230.9	255.255.255.252
6		Fa1/0	172.16.226.11	255.255.254.0
7	RTC	Se2/0	172.16.230.6	255.255.255.252
8		Se3/0	172.16.230.13	255.255.255.252
9		Fa0/0	172.16.230.10	255.255.255.252
10		Fa1/0	172.16.228.11	255.255.255.0
10	RTD	Se2/0	172.16.230.14	255.255.255.252
11		Fa0/0	172.16.229.222	255.255.255.0

Step 2: Given the information in file, answer the following questions:

- How many different subnets are there? 8
- What is the network address of each of these subnets? (*Hint: Given the IP address and the subnet mask, you can calculate the network address using AND operation*). Label the subnets in the topology given in Figure 1, and complete Table 2 below.

Table 2

Subnet #	Network Address	Broadcast Address	Range of usable addresses
1	172.16.224.0	172.16.225.255	172.16.224.1 - 172.16.225.254
2	172.16.230.0	172.16.230.3	172.16.230.1 - 172.16.230.2
3	172.16.226.0	172.16.227.255	172.16.226.1-172.16.227.254
4	172.16.230.4	172.16.230.7	172.16.230.5-172.16.230.6
5	172.16.230.12	172.16.230.15	172.16.230.13-172.16.230.14
6	172.16.230.8	172.16.230.11	172.16.230.9-172.16.230.10
7	172.16.228.0	172.16.228.255	172.16.228.1-172.16.228.254
8	172.16.229.0	172.16.229.255	172.16.229.1-172.16.229.254

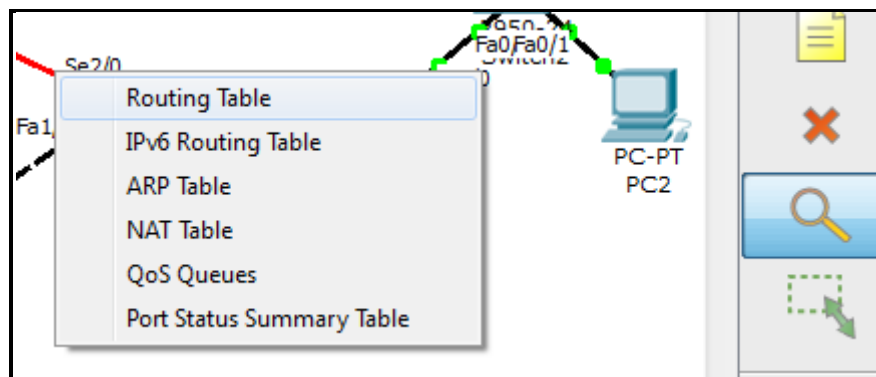
- Provided that all PC will use the last usable address in its subnet, fill in Table 3 below with the correct information.

Table 3

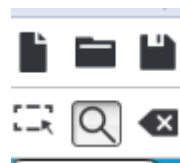
#	Device Name	IP Address	Subnet Mask	Default Gateway
1	PCA	172.16.225.254	255.255.254.0	172.16.224.255
2	PCB	172.16.227.254	255.255.254.0	172.16.226.11
3	PCC	172.16.228.254	255.255.255.0	172.16.228.11
4	PCD	172.16.229.25	255.255.255.0	172.16.229.222

Step 3: Complete the IP addressing information on all the PCs in the topology. (Hint: Click on the PC, choose the **Desktop** tab, then click **IP Configuration**).

Step 4: Open the routing table for each router. (Hint: you can use the 'magnifying glass' icon, then point to a router and choose 'Routing Table'. See Figure 3 below.)



(a)



(b)

Figure 3

Step 5: Copy the image of the routing table for each router. (Hint: You can use ‘Snipping Tool’ to copy the image.)

Routing Table for RTA

Type	Network	Port	Next Hop IP	Metric
C	172.16.224.0/23	FastEthernet0/0	---	0/0
C	172.16.230.0/30	Serial3/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0

Routing Table for RTB

Type	Network	Port	Next Hop IP	Metric
C	172.16.226.0/23	FastEthernet1/0	---	0/0
C	172.16.230.0/30	Serial2/0	---	0/0
C	172.16.230.8/30	FastEthernet0/0	---	0/0

Routing Table for RTC

Type	Network	Port	Next Hop IP	Metric
C	172.16.228.0/24	FastEthernet1/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0
C	172.16.230.8/30	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial3/0	---	0/0

Routing Table for RTD

Type	Network	Port	Next Hop IP	Metric
C	172.16.229.0/24	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial2/0	---	0/0

Step 6: Answer the questions below.

- a. Do all the routers have the same information in its routing table?

Ans : No.

- b. What is the difference that can be seen?

Ans : The network addresses are different from one and another.

- c. Can all the PCs ping each other successfully?

Ans :

#	Ping between devices	Successful (✓)	Unsuccessful (×)
1	PCA-PCB		(×)
2	PCA-PCC		(×)
3	PCA-PCD		(×)
4	PCB-PCC	(✓)	
5	PCB-PCD		(×)
6	PCC-PCD		(×)

- d. Reflection: what is the reason for your answer in (c)?

Ans : Only PCB and PCD ping each other successfully while others is unsuccessful because they cannot establish connections between each other.

Task 2: Dynamic routing configuration – RIP

Dynamic routing allows the network to be more flexible to changes. It can help the routers adapt to the changes in the pathways without much intervention from network administrators.

In this part of the lab, you will learn how to configure RIP routing protocol, and see how changes happen in the routing tables. Routers R1 and R4 is already configured for you.

Step 1: Choose Router RTA. Click the CLI tab. Copy the following text into the command line interface.

```
RTA>enable
RTA#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
RTA(config)#router rip
RTA(config-router)#version 2
RTA(config-router)#network 172.16.0.0
RTA(config-router)#no auto-summary
RTA(config-router)#exit
RTA(config)#exit
RTA#
%SYS-5-CONFIG_I: Configured from console by console
RTA#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
RTA#
```

Task 1.1:

(a) Copy (paste image) of the RTA routing table here.

Routing Table for RTA				
Type	Network	Port	Next Hop IP	Metric
C	172.16.224.0/23	FastEthernet0/0	---	0/0
C	172.16.230.0/30	Serial3/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0

Step 2: Choose Router RTB. Click the CLI tab. Copy the following text into the command line interface.

```
RTB>enable
RTB#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
RTB(config)#router rip
RTB(config-router)#version 2
RTB(config-router)#network 172.16.0.0
RTB(config-router)#no auto-summary
RTB(config-router)#exit
RTB(config)#exit
RTB#
%SYS-5-CONFIG_I: Configured from console by console
RTB#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
RTB#
```

Task 2.1:

- (a) Copy (paste image) of the RTB routing table and a new look into RTA routing table here.

RTB :

Routing Table for RTB					
Type	Network	Port	Next Hop IP	Metric	
R	172.16.224.0/23	Serial2/0	172.16.230.1	120/1	
C	172.16.226.0/23	FastEthernet1/0	---	0/0	
C	172.16.230.0/30	Serial2/0	---	0/0	
R	172.16.230.4/30	Serial2/0	172.16.230.1	120/1	
C	172.16.230.8/30	FastEthernet0/0	---	0/0	

New look RTA:

Routing Table for RTA					
Type	Network	Port	Next Hop IP	Metric	
C	172.16.224.0/23	FastEthernet0/0	---	0/0	
R	172.16.226.0/23	Serial3/0	172.16.230.2	120/1	
C	172.16.230.0/30	Serial3/0	---	0/0	
C	172.16.230.4/30	Serial2/0	---	0/0	
R	172.16.230.8/30	Serial3/0	172.16.230.2	120/1	

- (b) Reflection: what difference do you see between routing tables of RTA and RTB?

Ans : The difference between the routing table are on the Next Hop IP. This is because the IP addresses are different from each other. However, their network address does not change.

Step 3: Copy the same configuration instructions to RTC and RTD

RTC :

Routing Table for RTC				
Type	Network	Port	Next Hop IP	Metric
R	172.16.224.0/23	Serial2/0	172.16.230.5	120/1
R	172.16.226.0/23	FastEthernet0/0	172.16.230.9	120/1
C	172.16.228.0/24	FastEthernet1/0	---	0/0
R	172.16.230.0/30	Serial2/0	172.16.230.5	120/1
R	172.16.230.0/30	FastEthernet0/0	172.16.230.9	120/1
C	172.16.230.4/30	Serial2/0	---	0/0
C	172.16.230.8/30	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial3/0	---	0/0

RTD :

Routing Table for RTD				
Type	Network	Port	Next Hop IP	Metric
R	172.16.224.0/23	Serial2/0	172.16.230.13	120/2
R	172.16.226.0/23	Serial2/0	172.16.230.13	120/2
R	172.16.228.0/24	Serial2/0	172.16.230.13	120/1
C	172.16.229.0/24	FastEthernet0/0	---	0/0
R	172.16.230.0/30	Serial2/0	172.16.230.13	120/2
R	172.16.230.4/30	Serial2/0	172.16.230.13	120/1
R	172.16.230.8/30	Serial2/0	172.16.230.13	120/1
C	172.16.230.12/30	Serial2/0	---	0/0

Step 4: Answer the questions below.

- a. Do all the routers have the same information in its routing table?

Ans : No.

- b. Write down what RTC and RTD routing table information (Next Hop IP, Metric) to the network 172.16.224.0/24.

Ans : RTC next hop IP is 172.16.230.5 and its metric is 120/1 whereas RTD next hop IP is 172.16.230.13 and its metric is 120/2.

- c. What is the difference that can be seen? Why is this?

Ans : The Next Hop Ip and the metric are different. This is because RTC used FastEthernet 1/0 while RTD used FastEthernet 0/0.

- d. Can all the PCs ping each other successfully?

Table 5

#	Ping between devices	Successful (✓)	Unsuccessful (✗)
1	PCA-PCB	(✓)	
2	PCA-PCC	(✓)	
3	PCA-PCD	(✓)	
4	PCB-PCC	(✓)	
5	PCB-PCD		(✗)
6	PCC-PCD	(✓)	

- e. Reflection: what is the reason for your answer in (d)?

Ans : Some routers are not configured properly via the switches so they are not connected properly to each other. It makes the outcome to be one of it cannot ping correctly.

Step 5: Switch off router RTA. What are the changes noted in the routing tables?

Ans : The routing table is blank.

Step 6: Switch on router RTA. What are the changes noted in the routing tables?

Ans : All the routing table data is displayed back again.

Step 7: Reflection: What have you learned in this task?

Ans : All router configuration is saved in NVRAM. When there is no power, settings are preserved in NVRAM.

Task 3: Dynamic routing configuration – OSPF

Make sure that you have all the routing tables on display on one side (as before). As you go through the steps, look at the changes happening in the routing tables.

Step 1: On **all** the routers, do the following.

```
Router#configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#no router rip  
Router(config)#exit  
Router#  
%SYS-5-CONFIG_I: Configured from console by console  
Router#copy running-config startup-config  
Destination filename [startup-config]?
```

Step 2: Copy the image of the routing table for each router.

Routing Table for RTA

Type	Network	Port	Next Hop IP	Metric
C	172.16.224.0/23	FastEthernet0/0	---	0/0
C	172.16.230.0/30	Serial3/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0

Routing Table for RTB

Type	Network	Port	Next Hop IP	Metric
C	172.16.226.0/23	FastEthernet1/0	---	0/0
C	172.16.230.0/30	Serial2/0	---	0/0
C	172.16.230.8/30	FastEthernet0/0	---	0/0

Routing Table for RTC

Type	Network	Port	Next Hop IP	Metric
C	172.16.228.0/24	FastEthernet1/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0
C	172.16.230.8/30	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial3/0	---	0/0

Routing Table for RTD

Type	Network	Port	Next Hop IP	Metric
C	172.16.229.0/24	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial2/0	---	0/0

Step 3: For Router RTA, Click the CLI tab. Copy the following text into the command line interface.

```
RTA# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
RTA(config)#router ospf 1
RTA(config-router)#network 172.16.224.0 0.0.1.255 area 0
RTA(config-router)#network 172.16.230.0 0.0.0.3 area 0
RTA(config-router)#network 172.16.230.4 0.0.0.3 area 0
RTA(config-router)#end
RTA# copy running-config startup-config
Destination filename [startup-config]?
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
RTA#
```

Task 3.1: paste the image of RTA's routing table here.

Routing Table for RTA				
Type	Network	Port	Next Hop IP	Metric
C	172.16.224.0/23	FastEthernet0/0	---	0/0
C	172.16.230.0/30	Serial3/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0

Task 3.2:

- Does RTA have a path to ALL the different subnet?
Ans : No.
- Try pinging the different PCs and jot down your results.

Table 6

#	Ping between devices	Successful (✓)	Unsuccessful (x)
1	PCA-PCB		(x)
2	PCA-PCC		(x)
3	PCA-PCD		(x)

Step 4: Configure the other routers with OSPF routing algorithm.

Step 4.1: For Router RTB, Click the CLI tab. Copy the following text into the command line interface.

```
RTB# configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
RTB(config)#router ospf 1  
RTB(config-router)#network 172.16.226.0 0.0.1.255 area 0  
RTB(config-router)#network 172.16.230.0 0.0.0.3 area 0  
RTB(config-router)#network 172.16.230.8 0.0.0.3 area 0  
RTB(config-router)#end  
RTB# copy running-config startup-config  
Destination filename [startup-config]?  
%SYS-5-CONFIG_I: Configured from console by console  
Building configuration...  
[OK]  
RTB#
```

Step 4.2: For Router RTC, Click the CLI tab. Copy the following text into the command line interface.

```
RTC# configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
RTC(config)#router ospf 1  
RTC(config-router)#network 172.16.228.0 0.0.0.255 area 0  
RTC(config-router)#network 172.16.230.4 0.0.0.3 area 0  
RTC(config-router)#network 172.16.230.8 0.0.0.3 area 0  
RTC(config-router)#network 172.16.230.12 0.0.0.3 area 0  
RTC(config-router)#end  
RTC# copy running-config startup-config  
Destination filename [startup-config]?  
%SYS-5-CONFIG_I: Configured from console by console  
Building configuration...  
[OK]  
RTC#
```

Step 4.3: For Router RTD, Click the CLI tab. Copy the following text into the command line interface.

```
RTD# configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
RTD(config)#  
RTD(config)#router ospf 1  
RTD(config-router)#network 172.16.229.0 0.0.0.255 area 0  
RTD(config-router)#network 172.16.230.12 0.0.0.3 area 0  
RTD(config-router)#end  
RTD# copy running-config startup-config  
Destination filename [startup-config]?  
Building configuration...  
[OK]  
RTD#  
%SYS-5-CONFIG_I: Configured from console by console  
RTD#
```

Step 5: Copy the image of the routing table for each router and paste it here.

Routing Table for RTA

Type	Network	Port	Next Hop IP	Metric
C	172.16.224.0/23	FastEthernet0/0	---	0/0
O	172.16.226.0/23	Serial3/0	172.16.230.2	110/65
O	172.16.228.0/24	Serial2/0	172.16.230.6	110/65
O	172.16.229.0/24	Serial2/0	172.16.230.6	110/129
C	172.16.230.0/30	Serial3/0	---	0/0
C	172.16.230.4/30	Serial2/0	---	0/0
O	172.16.230.8/30	Serial3/0	172.16.230.2	110/65
O	172.16.230.8/30	Serial2/0	172.16.230.6	110/65
O	172.16.230.12/30	Serial2/0	172.16.230.6	110/128

Routing Table for RTB

Type	Network	Port	Next Hop IP	Metric
O	172.16.224.0/23	Serial2/0	172.16.230.1	110/65
C	172.16.226.0/23	FastEthernet1/0	---	0/0
O	172.16.228.0/24	FastEthernet0/0	172.16.230.10	110/2
O	172.16.229.0/24	FastEthernet0/0	172.16.230.10	110/66
C	172.16.230.0/30	Serial2/0	---	0/0
O	172.16.230.4/30	FastEthernet0/0	172.16.230.10	110/65
C	172.16.230.8/30	FastEthernet0/0	---	0/0
O	172.16.230.12/30	FastEthernet0/0	172.16.230.10	110/65

Routing Table for RTC

Type	Network	Port	Next Hop IP	Metric
O	172.16.224.0/23	Serial2/0	172.16.230.5	110/65
O	172.16.226.0/23	FastEthernet0/0	172.16.230.9	110/2
C	172.16.228.0/24	FastEthernet1/0	---	0/0
O	172.16.229.0/24	Serial3/0	172.16.230.14	110/65
O	172.16.230.0/30	FastEthernet0/0	172.16.230.9	110/65
C	172.16.230.4/30	Serial2/0	---	0/0
C	172.16.230.8/30	FastEthernet0/0	---	0/0
C	172.16.230.12/30	Serial3/0	---	0/0

Routing Table for RTD

Type	Network	Port	Next Hop IP	Metric
O	172.16.224.0/23	Serial2/0	172.16.230.13	110/129
O	172.16.226.0/23	Serial2/0	172.16.230.13	110/66
O	172.16.228.0/24	Serial2/0	172.16.230.13	110/65
C	172.16.229.0/24	FastEthernet0/0	---	0/0
O	172.16.230.0/30	Serial2/0	172.16.230.13	110/129
O	172.16.230.4/30	Serial2/0	172.16.230.13	110/128
O	172.16.230.8/30	Serial2/0	172.16.230.13	110/65
C	172.16.230.12/30	Serial2/0	---	0/0

Step 6: Switch off router RTA. What are the changes noted in the routing tables?

Ans : The routing table is blank.

Step 7: Switch on router RTA. Wait a few minutes. What are the changes noted in the routing tables?

Ans : All the routing table data is displayed back again.

Step 8: Reflection: What have you learned in this task?

Ans : In this task, I have learned configuration of OSPF restore the connection to its previous state without requiring any new settings if one of the routers is turned off

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